

Reviewer Pack — Minimal Local Index of Configuration Spaces vs Sipser Functi...

Entry 08610d2777ba · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-05-25 11:43:18 UTC

Minimal Local Index of Configuration Spaces vs Sipser Function Entropic Complexity

Entry ID: 08610d2777ba **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-25 11:43:18 UTC

1. Conjecture

Field A (mathematical branch): Configuration Space Theory **Field B** (complexity object): Complexity Theory: Entropy-based Communication Complexity

Statement:

{'text': 'For every explicit function f in P with a Sipser function representation, the minimal local index of its configuration space is upper bounded by the entropy-based communication complexity of f .', 'quantitative_relation': 'E[MinimalLocalIndex(f)] = O(EntropyBasedCommunicationComplexity(f))'}

Rationale (proposer's reasoning):

{'text': 'Configuration spaces provide a geometric perspective on complexity classes, and their local index could reveal hidden structure that is not evident from classical circuit-based or communication-complexity measures.', 'why_structure_exposed': 'A direct connection between the geometric invariant and entropic complexity might expose new insights into the nature of computation.'}

Taxonomy category: ACC_SIPSER (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): e7d76ab94f571871

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if the mean minimal local index is less than or equal to three times the entropy-based communication complexity, across all seeds.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 2 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "configuration space theory" AND "Sipser function entropic complexity"
- "minimal local index" AND "entropy-based communication complexity" - "P class functions"
AND minimal local index upper bound entropy-based communication complexity

Top relevant hits considered: - [http://arxiv.org/abs/2001.06906v1] Some inequalities for P-class functions - [s2:a6ae230cef877cad863a11d41c6234830c905733] Algorithms and computation : 16th International Symposium, ISAAC 2005, Sanya, Hainan, China, December 19-21, 2005 : proc

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=0, elapsed=0.0s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    n = 10 # Number of instances per trial
    results = []

    for _ in range(n):
        f = [random.randint(0, 1) for _ in range(10)] # Generate a random binary function

        # Calculate the minimal local index (simplified example)
        min_local_index = sum(f)

        # Calculate the entropy-based communication complexity (simplified example)
        entropy_based_comm_complexity = sum([f.count(x) * math.log2(f.count(x)) for x in set(f)])

        results.append({
            "metric_name": "MinimalLocalIndex",
            "metric_value": min_local_index,
            "instances_tested": n,
            "conjecture_holds": min_local_index <= 3 * entropy_based_comm_complexity,
            "counterexample": "" if min_local_index <= 3 * entropy_based_comm_complexity else f"Co
        })

    return {
        "seed": seed,
        "metric_name": "MinimalLocalIndex",
        "metric_value": sum(result["metric_value"] for result in results) / n,
```

```

        "instances_tested": n,
        "conjecture_holds": all(result["conjecture_holds"] for result in results),
        "counterexample": "" if all(result["conjecture_holds"] for result in results) else next((r
    }

if __name__ == "__main__":
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3

    results = []

    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std=0.0 support_fraction=1.0")
    elif any(not r["conjecture_holds"] for r in results):
        first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
        print(f"RESULT: FALSIFIED counterexample=\"{results[0]['counterexample']}\" first_failing={first_failing_seed}")
    else:
        print(f"RESULT: INCONCLUSIVE reason=unknown")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

'metric_name': 'MinimalLocalIndex', 'metric_value': 5.0, 'instances_tested': 10, 'conjecture_holds': 10
TRIAL: {'seed': 421, 'metric_name': 'MinimalLocalIndex', 'metric_value': 4.9, 'instances_tested': 10, 'conjecture_holds': 10}
TRIAL: {'seed': 463, 'metric_name': 'MinimalLocalIndex', 'metric_value': 5.2, 'instances_tested': 10, 'conjecture_holds': 10}
TRIAL: {'seed': 503, 'metric_name': 'MinimalLocalIndex', 'metric_value': 4.6, 'instances_tested': 10, 'conjecture_holds': 10}
TRIAL: {'seed': 547, 'metric_name': 'MinimalLocalIndex', 'metric_value': 5.7, 'instances_tested': 10, 'conjecture_holds': 10}
TRIAL: {'seed': 593, 'metric_name': 'MinimalLocalIndex', 'metric_value': 4.8, 'instances_tested': 10, 'conjecture_holds': 10}
TRIAL: {'seed': 631, 'metric_name': 'MinimalLocalIndex', 'metric_value': 5.2, 'instances_tested': 10, 'conjecture_holds': 10}
TRIAL: {'seed': 677, 'metric_name': 'MinimalLocalIndex', 'metric_value': 5.6, 'instances_tested': 10, 'conjecture_holds': 10}
TRIAL: {'seed': 727, 'metric_name': 'MinimalLocalIndex', 'metric_value': 4.0, 'instances_tested': 10, 'conjecture_holds': 10}
TRIAL: {'seed': 773, 'metric_name': 'MinimalLocalIndex', 'metric_value': 4.9, 'instances_tested': 10, 'conjecture_holds': 10}
TRIAL: {'seed': 821, 'metric_name': 'MinimalLocalIndex', 'metric_value': 5.9, 'instances_tested': 10, 'conjecture_holds': 10}
TRIAL: {'seed': 877, 'metric_name': 'MinimalLocalIndex', 'metric_value': 5.2, 'instances_tested': 10, 'conjecture_holds': 10}
TRIAL: {'seed': 929, 'metric_name': 'MinimalLocalIndex', 'metric_value': 5.6, 'instances_tested': 10, 'conjecture_holds': 10}
RESULT: SUPPORTED mean=4.923333333333333 std=0.0 support_fraction=1.0

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

The empirical test only includes a small number of instances ($n \leq 15$) and does not scale trivially with n . This raises concerns about the generalizability of the results. The metric ‘MinimalLocalIndex’ might not hold for larger values of n , indicating that the conjecture could be false for functions beyond those tested.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge | original judge: The mean MinimalLocalIndex across all seeds is 4.923333333333333, which is less than or equal to three times the mean EntropyBasedCommunicationComplex | next: Further investigation with a larger sample size and different functions is recommended to validate the conjecture’s generalizability.

11. Audit log (LLM calls)

Total LLM calls: 12

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	13273
2	propose	ollama_remote	glm4:latest	0	0	12004
3	propose	ollama_remote	glm4:latest	0	0	10021
4	preregistration	ollama_remote	glm4:latest	0	0	5170
5	novelty	ollama_remote	glm4:latest	0	0	4590
6	novelty	ollama_remote	glm4:latest	0	0	9589
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10780
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	6559
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	6831
10	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7595
11	critic	ollama_remote	glm4:latest	0	0	9376
12	judge	ollama_remote	glm4:latest	0	0	5753

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 101541 ms total latency. Provider mix: {'ollama_remote': 12}

(full prompt+response transcripts available in `research/audit/08610d2777ba.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/08610d2777ba.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/08610d2777ba.tar.gz` (if generated)